BIP-39 Mapping Explained

Step 1: Generate Entropy
- 128 bits (for 12 words), 192 bits (for 18 words), or 256 bits (for 24 words).

Step 2: Add Checksum
- Compute SHA-256 of entropy.
- Take the first (ENT / 32) bits as checksum.
- Append checksum to entropy.
- Example: 128 bits entropy + 4 checksum bits = 132 bits total.

Step 3: Split into 11-bit chunks
- Each BIP-39 word represents 11 bits (since 2^11 = 2048).
- 132 bits ÷ 11 = 12 words.
- 264 bits ÷ 11 = 24 words.
- Result: N × 11-bit chunks.

Step 4: Convert each 11-bit chunk to an index
- Each chunk is a number 0–2047.
- Example:
00000000000 → 0
00000000001 → 1
11111111111 → 2047

Step 5: Map index → word
- Each index corresponds to a word in the BIP-39 wordlist.
- Wordlist length = 2048 words.
- Example:
Index 0 → abandon
Index 2047 → zoo

Step 6: Final Seed Phrase
- Sequence of 12 / 18 / 24 words.
- Last word contains checksum bits, so changing entropy alters the last word.
- Invalid phrases are detected when checksum doesn’t match entropy.